Human Data

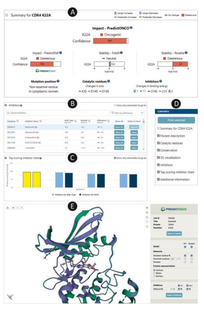
ELIXIR CZECH REPUBLIC

Historically, ELIXIR CZ focused on bioinformatics solutions for precision medicine—addressing haematological, oncological, autoimmune, and neurodegenerative diseases—which recently led to the development of ML-driven tools such as PredictOnco. However, with the scientific community's push for open science and AI-driven healthcare, our focus is now shifting toward establishing a federated human data infrastructure. This new direction aims to integrate and standardize sensitive human omics data across Europe, enabling secure, interoperable data sharing and analysis that will drive AI model training and innovative diagnostics. By aligning with initiatives such as GDI, FEGA and EOSC-CZ, ELIXIR CZ is committed to developing a robust national and pan-European platform that ultimately enhances personalized medicine through data-driven insights and open science collaboration.

Precision medicine

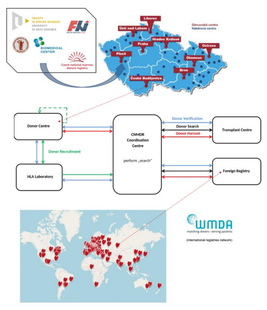
PREDICTONCO

- Estimates mutation impact on patient health using advanced AI-driven analytics
- Developed in collaboration with BioVendor for seamless integration with hospital systems



MDRsim: Tool for Bone Marrow Donor Search

- Optimizes donor matching using advanced simulation and predictive models
- Developed in collaboration with Sampling Human to integrate cutting-edge AI



ELIXIR Communities

- Main connection: Federated Human Data
  - metadata models, security and authentication, TREs, computational workflows
- Also involved in: Human Copy Number Variation, Cancer Data, Rare Diseases, and Galaxy

GDI

European Genomic Data Infrastructure

- Incorporation of GDI-defined standards
- Key ELIXIR CZ involvement in
  - federated authentication (LS AAI)
  - federated compute (Kubernetes)

Data Access Management

Data Processing

FEGA

Federated European Genome-phenome Archive

- Essential for CZ to become an FEGA node, as it is the most developed Federated Human Data platform
- ELIXIR CZ actively supports the transition, leveraging collaboration with more mature members (e.g., Switzerland) SIB

EOSC-CZ

National Czech Programme | eosc

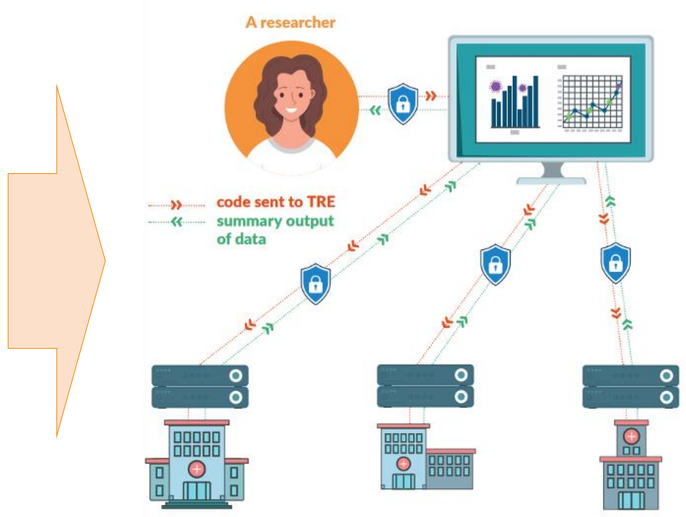
- Encompasses three large-scale open science projects, funded by the Czech government and EU, that drive national data infrastructure development.
- The National OMICS Data Platform stands as a key repository in the Bio/Health/Food workgroup

Bio/Health/Food

Strategic goals

- Establish a national omics repository as an integrated node in a forming European Federated Human Data network, ensuring a scalable, secure platform for diverse omics datasets
- Integrated within the Beacon network and connected as a FEGA node
- Incentivize data producers via seamless uploading and primary data analysis
- Ensure connectivity with clinical repositories, emphasizing clinical and pharma development applications through a dedicated, MD-friendly interface
- Plan to integrate a feedback loop between data producers, clinicians, and researchers to continuously improve data quality and utilization

- Enable federated compute for scalable analytics
- Support data visiting paradigm by returning anonymized, aggregated results from sensitive data processing through verified workflows



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